Amines

Amine

An amine is a substituted ammonia molecule with basic properties and has the general formula RNH_2 , R_2NH , R_3N where R is an alkyl or an aryl group.

NB amines are classified by the class of the <u>nitrogen</u>, primary amines have one carbon bonded to N, secondary amines have two carbons attached directly to the N, etc.

Nomenclature.

Common aliphatic amines are named as **"alkyl amines"**

CH_3NH_2	$(CH_3)_2NH$	$(CH_3)_3N$
methylamine	dimethylamine	trimethylamine
1^o	2^o	3^o

$CH_3CH_2NHCH_3$

ethylmethylamine

2^o

$CH_3CH_2CHCH_3$

$\quad\quad\quad\quad\;|$

$\quad\quad\quad\; NH_2$

sec-butylamine

1^o

$\quad\quad\; CH_3$

$\quad\quad\quad |$

CH_3CCH_3

$\quad\quad\quad |$

$\quad\quad\; NH_2$

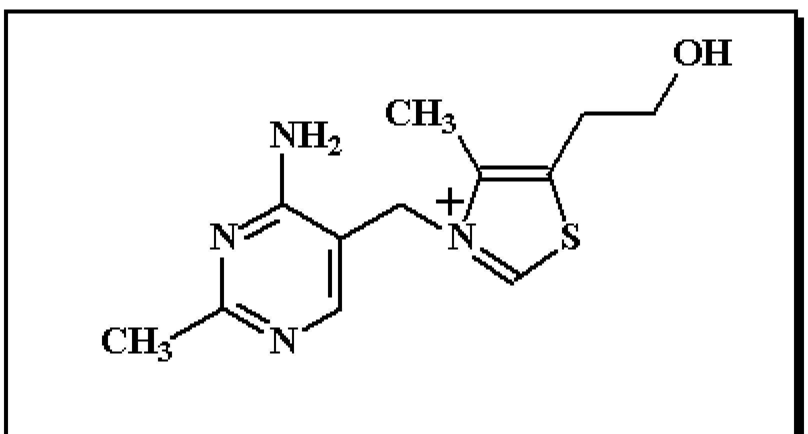
tert-butylamine

1^o

Sources and uses of amines

Nitrogen compounds are found throughout the plant and animal, substituted amine, and amide occur in every living cell. Many of these compounds have important physiological activity for examples many antibacterial agents contain nitrogen. Common examples include the synthetic sulfa drugs and penicillin-related antibiotics which are synthesized by molds

Amines and amides are part of the structures of the B-complex vitamins examples are thiamine (vitamin B1) and nicotinamide (niacin)



Thiamine

Heterocyclic amine & Risk

Some heterocyclic amines (HCAs) found in cooked and especially burned meat are known carcinogens. Research has shown that heterocyclic amine formation in meat occurs at high cooking temperatures.

For example; heterocyclic amines are the carcinogenic chemicals formed from the cooking of muscle meats such as beef, pork, and fish.

Heterocyclic amines HCAs form when amino acids and creatine (achemical found in muscles) react at high cooking temperatures. Researchers have identified 17 different HCAs resulting from the cooking of muscle meats that may pose human cancer risk.

Amides

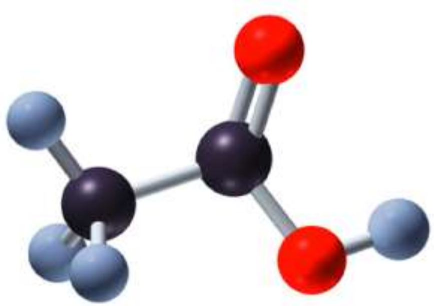
- **In amides, an amino group replaces the –OH group of carboxylic acids.**

$$CH_3-\overset{\overset{\displaystyle O}{\|}}{C}-OH$$

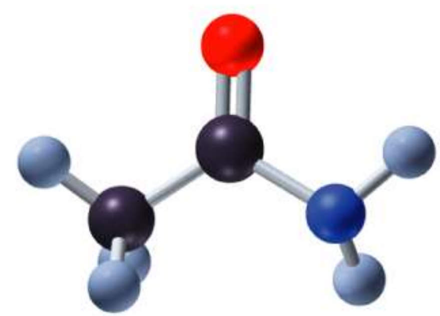
Carboxylic acid

$$CH_3-\overset{\overset{\displaystyle O}{\|}}{C}-NH_2$$

Amide



Ethanoic acid
(Acetic acid)



Ethanamide
(Acetamide)

Preparation of Amides

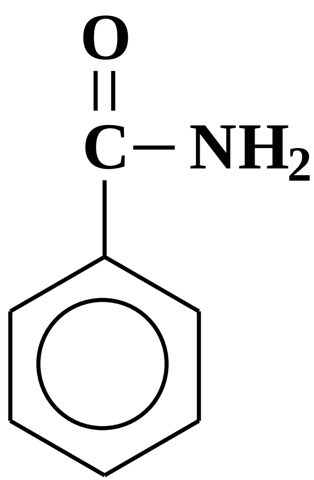
- **Amides are produced by reacting a carboxylic acid with ammonia or an amine (1° or 2°).**

$$CH_3-\overset{\displaystyle O}{\overset{\|}{C}}-OH + NH_3 \longrightarrow CH_3-\overset{\displaystyle O}{\overset{\|}{C}}-NH_2 + H_2O$$

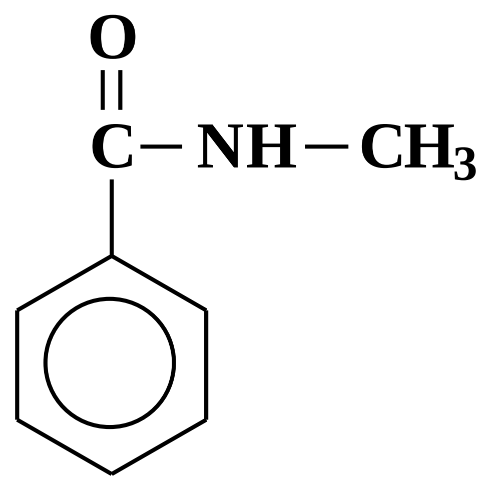
$$CH_3-\overset{\displaystyle O}{\overset{\|}{C}}-OH + CH_3-NH_2 \longrightarrow CH_3-\overset{\displaystyle O}{\overset{\|}{C}}-NH-CH_3 + H_2O$$

Aromatic Amides

- **The amide of benzene is named benzamide.**



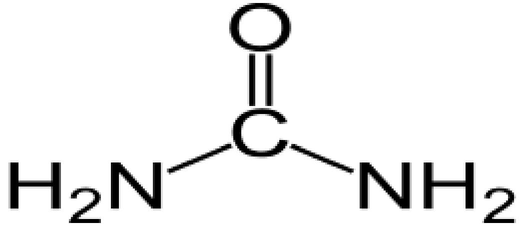
Benzamide

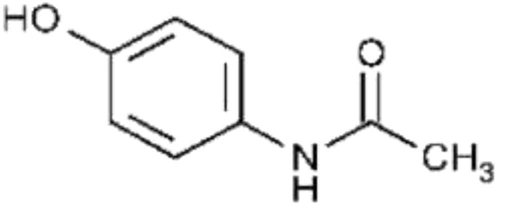


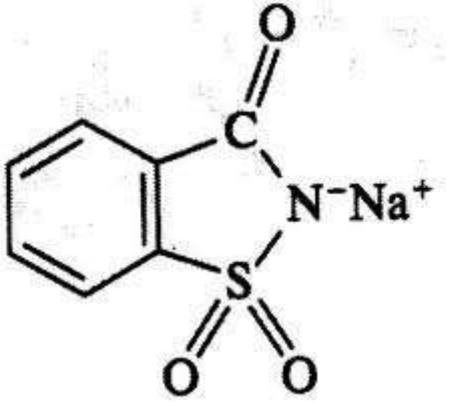
***N*-methylbenzamide**

Amides in Health and Medicine

- **Urea is the end product of protein metabolism.**
- **Saccharin is an artificial sweetener.**
- **Acetaminophen is used to reduce fever and pain.**

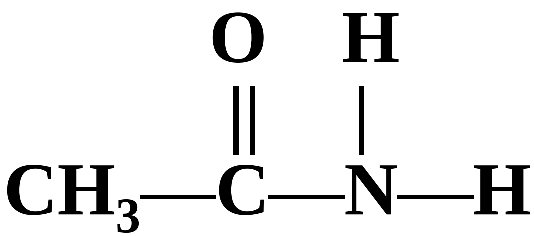




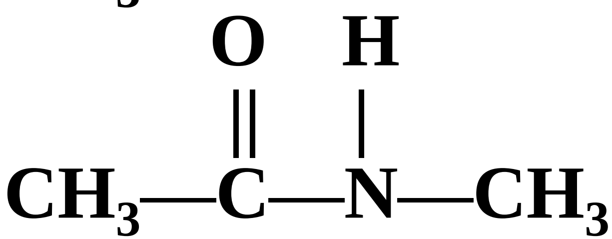


Classification of Amides

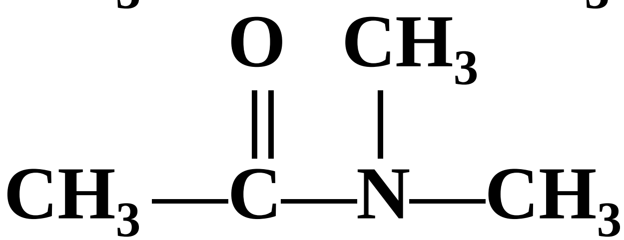
- Amides are classified according to the number of carbon atoms bonded to the nitrogen atom.



Primary (1°) amide



Secondary (2°) amide

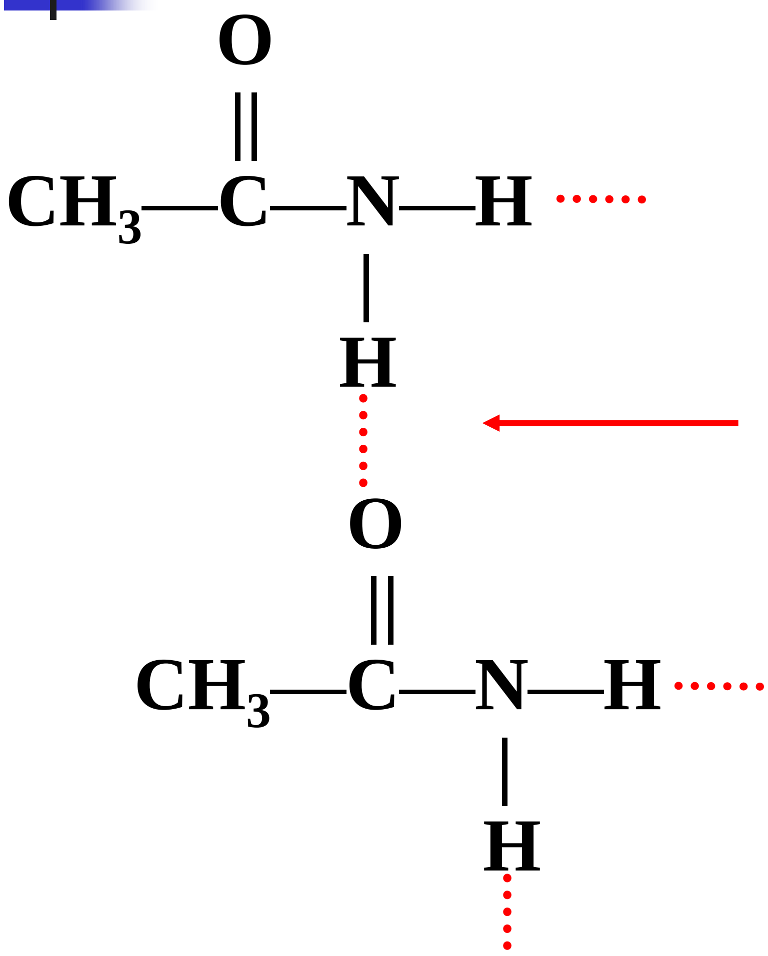


Tertiary (3°) amide

Physical Properties of Amides

- **Hydrogen bonds form in primary and secondary amides, but not tertiary amides.**
- **The melting points of primary amides are higher than secondary amides, which have higher melting points than tertiary amides.**
- **All amides can form hydrogen bonds with water.**
- **Amides with 1-5 carbon atoms are soluble in water.**

Hydrogen Bonding of Amides



Hydrogen bonding occurs between primary amides.

Reactions of Amides

Amides undergo:

- **Acid hydrolysis to produce a carboxylic acid and ammonium salt.**

- **Base hydrolysis to produce the salt of a carboxylic acid and an amine or ammonia.**

Hydrolysis Reactions